

Meridian Performance Galaxy - Product Requirements Document (PRD)

Version: 1.0

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Product Owner: Meridian Product Team

Status: Active Development

Classification: Internal / Academic

1. Document Purpose

This Product Requirements Document (PRD) defines the functional and non-functional requirements for the Meridian Performance Galaxy platform. It serves as the primary reference for design, engineering, and QA teams, as well as for academic evaluation of the product's scope and technical ambition.

2. Product Overview

2.1 Product Vision

Meridian Performance Galaxy reimagines how organizations understand employee performance. Instead of tables and bar charts, Meridian presents the entire workforce as an animated, interactive galaxy — where each star is an employee, and its visual properties encode key performance attributes in real time.

2.2 Product Goals

1. Reduce time-to-insight for HR and executive stakeholders from hours to seconds
2. Encode at least 5 performance dimensions simultaneously in a single visual
3. Support organizations of 10 to 10,000 employees without performance degradation
4. Require zero training for first-time users to navigate the galaxy

2.3 Out of Scope (v1.0)

- Mobile native applications (iOS/Android)
 - Real-time data sync via live HRIS API
 - AI-generated performance recommendations
 - Multi-language UI (planned for v1.2)
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3. User Stories

3.1 HR Director

ID	User Story	Priority
US-01	As an HR Director, I want to see all employees visualized on a single screen so I can quickly identify top and low performers	P0
US-02	As an HR Director, I want to filter employees by department so I can assess team-level performance	P0
US-03	As an HR Director, I want to export a screenshot of the galaxy to include in board presentations	P1
US-04	As an HR Director, I want to search for a specific employee by name or role	P0

3.2 Executive

ID	User Story	Priority
US-05	As a CEO, I want to see the bonus pool and goal hit rate at a glance in the header	P0
US-06	As a CFO, I want to sort the leaderboard by bonus to understand cost distribution	P1

3.3 Team Lead

ID	User Story	Priority
US-07	As a Team Lead, I want to click on a star to see an employee's full performance profile	P0
US-08	As a Team Lead, I want to enable orbit trails to visualize which employees are moving fastest	P2
US-09	As a Team Lead, I want to control orbit speed to slow down or speed up the simulation	P2

4. Functional Requirements

4.1 Galaxy Visualization Engine

ID	Requirement	Priority
FR-01	The system shall render each employee as a circular star on an elliptical orbit around a central sun	P0
FR-02	Star size shall be proportional to the employee's performance score (range: 70-100)	P0
FR-03	Star color shall correspond to the employee's department using a fixed color palette	P0
FR-04	Orbit radius shall be proportional to the employee's tenure in years	P0
FR-05	Stars belonging to "Excellence" employees shall display a gold outer ring	P0
FR-06	Excellence employees shall be connected by animated dashed lines (Excellence Links)	P1
FR-07	The galaxy shall animate continuously at approximately 60fps	P0
FR-08	Stars shall emit a colored glow effect matching their department color	P1
FR-09	Background shall include a starfield of 300+ randomly positioned static stars with subtle twinkling	P2
FR-10	The central sun shall be rendered with multi-layer radial gradient glow effects	P1

4.2 Interaction & Navigation

ID	Requirement	Priority
FR-11	Users shall be able to drag horizontally to rotate (orbit) the galaxy	P0
FR-12	Users shall be able to scroll (mouse wheel) to zoom in/out (range: 0.4x - 2.5x)	P0
FR-13	Hovering over a star shall highlight it with increased radius and display a tooltip	P0
FR-14	Clicking a star shall open an Employee Detail Panel on the right side	P0

ID	Requirement	Priority
FR-15	Clicking a star shall animate the camera to bring that star to the center of view	P1
FR-16	Clicking a selected star again shall deselect it and close the Detail Panel	P0

4.3 Header & KPI Bar

ID	Requirement	Priority
FR-17	The header shall display: Headcount, Avg Score (with YoY), Excellence count, Bonus Pool (with YoY), Goal Hit %	P0
FR-18	The header shall remain fixed at the top and not scroll	P0
FR-19	KPI values shall be clearly legible at all standard screen sizes (1280px and above)	P0

4.4 Leaderboard Panel

ID	Requirement	Priority
FR-20	The leaderboard shall list all employees ranked by the selected sort criterion	P0
FR-21	Leaderboard shall support sorting by: Score, Tenure, Bonus	P1
FR-22	Each leaderboard row shall display: rank, department color dot, name, role, excellence star, and value	P0
FR-23	Hovering a leaderboard row shall highlight the corresponding star in the galaxy	P0
FR-24	Clicking a leaderboard row shall select the employee and open the Detail Panel	P0
FR-25	The leaderboard shall be scrollable for organizations with more than 8 employees	P0

4.5 Employee Detail Panel

ID	Requirement	Priority
FR-	The detail panel shall display: avatar (initials), name, role, department, score, tenure,	P0

ID	Requirement	Priority
26	bonus, excellence status, rank	
FR-27	The panel shall include a visual performance bar showing score as a percentage	P0
FR-28	The panel shall include a close (×) button to dismiss	P0
FR-29	The panel shall animate open/close	P2

4.6 Search

ID	Requirement	Priority
FR-30	The search bar shall filter stars in real time as the user types	P0
FR-31	Search shall match against: employee name, role, and department	P0
FR-32	Non-matching stars shall be visually dimmed (opacity reduced)	P0
FR-33	Non-matching leaderboard rows shall be visually dimmed	P1
FR-34	A clear (×) button shall appear when a query is active and reset the search on click	P1

4.7 Department Filters

ID	Requirement	Priority
FR-35	Filter buttons shall be auto-generated from the dataset's unique departments	P0
FR-36	Selecting a department filter shall dim all stars and orbit rings from other departments	P0
FR-37	The "All" filter shall restore full visibility	P0
FR-38	Each filter button shall display the department name and count	P1

4.8 Sector Map (Minimap)

ID	Requirement	Priority
FR-39	A minimap shall display a top-down scaled overview of all star positions	P1

ID	Requirement	Priority
FR-40	The minimap shall update in real time in sync with the main galaxy	P0
FR-41	Selected employees shall be indicated with a white ring in the minimap	P1
FR-42	The current scan angle shall be displayed below the minimap	P2

4.9 Controls (Pause / Trails / Labels / Speed)

ID	Requirement	Priority
FR-43	A Pause/Play toggle shall stop and resume all orbital animation	P1
FR-44	A Trails toggle shall display the recent orbital path of each star	P2
FR-45	A Labels toggle shall show employee names permanently above all stars	P2
FR-46	A speed slider shall control orbit speed from 0x to 5x	P2
FR-47	An Export button shall download a PNG screenshot of the current canvas state	P1

4.10 Legend

ID	Requirement	Priority
FR-48	A legend panel shall explain the visual encoding: size=score, orbit=tenure, color=dept, ring=excellence, links=excellence connections	P0
FR-49	The legend shall be positioned to avoid overlapping key UI elements	P1

5. Non-Functional Requirements

5.1 Performance

ID	Requirement
NFR-01	The galaxy shall render at a minimum of 30fps on standard laptop hardware with up to 500 employees
NFR-02	Initial load time shall not exceed 3 seconds on a standard broadband connection

ID	Requirement
NFR-03	Canvas animation shall use requestAnimationFrame for optimal browser scheduling

5.2 Compatibility

ID	Requirement
NFR-04	The platform shall function on Chrome, Firefox, Safari, and Edge (latest 2 versions)
NFR-05	Minimum supported screen width: 1280px
NFR-06	No external dependencies required beyond a modern browser

5.3 Accessibility

ID	Requirement
NFR-07	All interactive elements shall have cursor:pointer feedback
NFR-08	Color shall not be the sole encoding for critical information (tooltips provide text backup)
NFR-09	Font sizes shall meet WCAG AA minimum contrast ratios for UI labels

5.4 Security & Privacy

ID	Requirement
NFR-10	No employee data shall be transmitted to external servers in the browser-only version
NFR-11	Future API-connected versions shall implement OAuth 2.0 authentication
NFR-12	PII (Personally Identifiable Information) fields shall be configurable to anonymize

6. Data Model

6.1 Employee Object Schema

json

```
{
  "id": "integer (unique)",
  "name": "string",
  "role": "string",
  "dept": "string (enum: Engineering | Design | Product | HR | Security | Data)",
  "score": "integer (0-100)",
  "tenure": "integer (years)",
  "excellence": "boolean",
  "bonus": "integer (M$K)"
}
```

6.2 Visual Encoding Map

Data Field	Visual Property	Encoding Formula
score	Star radius	$r = 6 + (score - 70) / 30 * 9$
tenure	Orbit radius	$orbit = (85 + tenure * 26) * zoom$
dept	Star color	Fixed palette per department
excellence	Gold ring	Boolean — ring displayed if true
excellence	Dashed links	Lines between all excellence employees

7. System Architecture

7.1 Technology Stack

Layer	Technology
Rendering	HTML5 Canvas 2D API
Logic	Vanilla JavaScript (ES6+)
Styling	CSS3 with custom properties
Hosting	Static (GitHub Pages / Vercel)
Future: Data	REST API / WebSocket for live HRIS sync

7.2 Component Structure

```
index.html
```


- └─ Header Component (KPIs + Controls)
- └─ Galaxy Canvas (main rendering loop)
- └─ Left Panel
 - └─ Leaderboard Component
 - └─ Sector Map / Minimap Component
- └─ Employee Detail Panel
- └─ Legend Panel
- └─ Filter Bar
- └─ Search Bar

8. Acceptance Criteria

Feature	Acceptance Criteria
Galaxy renders	All 12 employees visible as stars on first load
Hover tooltip	Tooltip appears within 100ms of hovering a star
Click to select	Detail panel opens with correct employee data
Search	Typing "eng" dims all non-Engineering employees
Export	PNG file downloads successfully with correct canvas content
Filter	Selecting "Design" dims all non-Design stars
Leaderboard sort	Switching to "Bonus" re-ranks list by bonus value
Pause	Animation freezes immediately on Pause click

9. Release Milestones

Milestone	Target Date	Deliverables
MVP (v0.1)	July 2025	Galaxy render, hover, click, search
Beta (v0.5)	September 2025	All P1 features, leaderboard sort, export
v1.0 Launch	November 2025	All P1+P2 features, performance optimization
v1.1	Q1 2026	HRIS API integration (BambooHR pilot)
v1.2	Q2 2026	Multi-language support, mobile responsive

10. Open Questions

#	Question	Owner	Status
OQ-01	Should orbit speed differ per employee or be uniform?	Product	Open
OQ-02	What is the maximum supported employee count for v1.0?	Engineering	Open
OQ-03	Should the Detail Panel support editing performance scores inline?	Product	Deferred to v1.1
OQ-04	Do we need role-based access control (manager vs exec views)?	Product	Deferred to v1.1

*This PRD is a living document and will be updated as product decisions evolve.
Prepared for academic and product planning purposes — Meridian Product Team, June 2025.*